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Finding Reliable Shortest Paths in Road Networks Under Uncertainty

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Abstract The aim of this study is to investigate the solution algorithm for solving the problem of determining reliable shortest paths in road networks with stochastic travel times. The availability of reliable shortest paths enables travelers, in the face of travel time uncertainty, to plan their trips with a pre-specified on-time arrival probability. In this study, the reliable shortest path between origin and destination nodes is determined using a multiple-criteria shortest path approach when link travel times follow normal distributions. The dominance conditions involved in such problems are established, thereby reducing the number of generated non-dominated paths during the search processes. Two solution algorithms, multi-criteria label-setting and A* algorithms, are proposed and their complexities analyzed. Computational results using large scale networks are presented. Numerical examples using data from a real-world advanced traveller information system is also given to illustrate the applicability of the solution algorithms in practice.

Keywords Reliable shortest path problem · Travel time reliability · Advanced traveller information system · Pre-trip planning application

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1 Introduction

1.1 Background

Information and communication technology advances have encouraged the development of advanced traveller information systems (ATIS) for alleviating traffic congestion, particularly in congested urban road networks. ATIS utilizes technological infrastructures to collect real-time traffic data which are then transmitted to a central traffic management centre to generate updated traffic information. Such generated traffic information is subsequently disseminated to road users so as to help them make better route choice decisions. Many case studies have found that the ATIS-based routing application is an effective tool for demand management in congested road networks; helping reduce system total travel time, conserve energy consumption and enhance traffic safety (Chen et al. 2007a, b; Li et al. 2012).

The ATIS-based routing application can be broadly classified as either an in-vehicle navigation system or a pre-trip planning system. The in-vehicle navigation system is generally installed in private cars to provide an en-route guidance service to drivers. On-board microprocessors and wireless communication devices are typically required by this system to calculate the optimal route and update traffic information during the journey. The recommended routes should be found and repeatedly updated for large-scale networks within a short time. In this situation, drivers of using such system may have less flexible to change their departure time, since their journeys generally have started when they are using such system. A pre-trip planning system is deployed at the traffic centre to provide a pre-trip planning service to the public through websites portals. Users generally request this pre-trip planning service before their journeys so as to simultaneously determine their optimal departure time and route choices. On occasions, a large number of users may request such services simultaneously. These requests require routes and schedules to be calculated within a reasonable time interval.

This study focuses on the development of the pre-trip planning system in large-scale congested road networks. Most existing pre-trip planning systems assume that link travel times are deterministic. However, link travel times in congested road networks are highly stochastic due to demand fluctuations and supply degradations (Shao et al. 2006; Chen et al. 2007a, b, 2011, 2012a; Tian et al. 2012). It has been found from empirical studies that travel time uncertainty has significant impacts on travellers' departure time and route choices (Lam and Small 2001; Tam et al. 2008; Wu and Nie 2011; Nie and Wu 2001). This is because greater travel time variations may cause late arrivals and impose high penalties on travellers (e.g. missed flights or lost business). To avoid such penalties, travellers tend to start trips earlier and choose reliable path to ensure a more certain on-time arrival probability (known, in the literature, as travel time reliability). Therefore, the sophisticated pre-trip planning system should allow travellers to incorporate travel time reliability concerns in their departure time and route choice decisions. In view of the above, this study aims to address the problem of finding the reliable shortest path

from an origin to a destination for the development of pre-trip planning system with consideration of travel time reliability.

1.2 Literature review

In the literature, much attention and effort have been devoted to incorporating the concept of travel time reliability into classical shortest path problems. For convenience, the one-to-one shortest path problem considering travel time reliability is hereafter referred as reliable shortest path problem (RSPP). Frank (1969) introduced the concept of finding the most reliable path which maximizes the probability of travellers arriving at their destination within a given travel time budget. Chen and Ji (2005) proposed a similar concept of finding the α-reliable path which minimizes the travel time budget required to ensure a pre-specified on-time arrival probability.

Based on the above two definitions, substantial solution algorithms have been proposed to solve RSPP in congested road networks. Mirchandani (1976) presented a recursive algorithm to solve the discrete version of Frank's problem. However, the algorithm requires path enumeration and thus cannot be applied to large-scale problems. Shao et al. (2004) proposed a heuristic method for solving this problem, based on the relationship between link travel time mean and standard deviation. Nonetheless, this method cannot provide a solution for risk-seeking travellers, whose travel time budget is less than the least expected travel time between origin–destination (O-D) nodes. Nikolova (2006) and (2010) proposed a parametric approach to determine both the α-reliable and most reliable paths. However, the proposed parametric approach also cannot determine the optimal solution for risk-seeking travellers. Chen and Ji (2005) presented a genetic algorithm for finding both the α-reliable and most reliable paths. It should be noted that the genetic algorithm is computationally expensive and the precision of results depends on the maximum number of simulations.

Miller-Hooks and Mahmassani (2003) noted that the shortest path problem in stochastic networks can be formulated and solved by multi-criteria shortest path approaches based on dominance conditions (e.g. first-order stochastic dominance (FSD)). Nie and Wu (2009) argued that the definition of α-reliable path is equivalent to that of most reliable path. Based on the FSD condition, Nie and Wu (2009) proposed a label-correcting algorithm to find both the α-reliable and most reliable paths by generating all non-dominated paths in the network. The proposed algorithm, notwithstanding, has a non-deterministic polynomial complexity, since the number of non-dominated paths under FSD grows exponentially with network size. To reduce the number of generated non-dominated paths, an approximate approach was proposed. Nevertheless, the approximation may miss the optimal paths completely, and there is no precise notion of how good that approximation is.

This study addresses RSPP in the context of pre-trip planning system. It is found in this study that the definition of α-reliable path is more suitable for the development of pre-trip planning system than that of most reliable path. The α-reliable path definition can not only better reflect travellers' route decision-making processes in pre-trip

planning systems, but also lead to stricter dominance conditions and more efficient solution algorithms. The paper differs significantly from previous work in the following aspects.

(1) Dominance conditions of RSPP are established based on the α-reliable path definition. These established conditions can determine dominated paths which may not be identified by FSD. Therefore, the established conditions can reduce the number of non-dominated paths generated, evaluated, and stored when searching for the α-reliable path, so as to improve the efficiency of the search process.
(2) Two multi-criteria solution algorithms are developed to exactly solve RSPP based on the α-reliable path definition. The proposed multi-criteria label-setting approach can determine the α-reliable path without the requirement of generating all non-dominated paths in the network. The proposed multi-criteria A* can further speed up the search process by assigning a higher priority to those nodes closer to the destination.
(3) The optimality and complexity of the proposed solution algorithms are proved theoretically, and the features and efficiency of the proposed model and solution algorithms are empirically demonstrated. Numerical results using data from a real-world ATIS show that the proposed solution algorithms are sufficiently robust to take account of travellers' various risk-taking behaviour, and are appropriate for developing large-scale pre-trip planning systems.

The rest of the paper is organized as follows. The definition of RSPP in relation to stochastic road networks, together with the model formulation is discussed in Section 2. The dominance conditions of RSPP are introduced in Section 3. Solution algorithms to solve RSPP are presented in Section 4. A case study using a real-world ATIS in Hong Kong is described in Section 5. The computational performance of the proposed solution algorithms is reported in Section 6. Finally, conclusions are given together with recommendations for further study.

2 Problem definition

Let G = (N, A) be a directed network, where N and A are the sets of nodes and links respectively. Suppose that the nodes $r \in N$ and $s \in N$ represent the O-D nodes. Each node i has a set of successor nodes $SCS(i) = \{j : a_{ij} \in A\}$ and a set of predecessor nodes $PDS(i) = \{k : a_{ki} \in A\}$. Assume that each link a_{ij} has a random travel time T_{ij} with a given probability density function (PDF). The mean and standard deviation (SD) of link travel time are denoted by t_{ij} and σ_{ij} respectively. A list of notations can be found in Appendix A.

Let $P^{rs} = \left\{p_1^{rs}, \ldots, p_n^{rs}\right\}$ be a set of paths from the origin r to the destination s. Let x_{ij}^{rs} be the decision variable regarding the link-path incidence relationship, where

$x_{ij}^{rs} = 1$ means that the link a_{ij} is on the path p_u^{rs}, and $x_{ij}^{rs} = 0$ otherwise. The path travel time, denoted by T_u^{rs}, is the sum of related link travel times along the path as

$$T_u^{rs} = \sum_{a_{ij} \in A} T_{ij} x_{ij}^{rs} \tag{1}$$

The path travel time T_u^{rs} is also a random variable whose distribution is the joint PDF of all link travel times along the path. The mean and SD of path travel time are denoted by t_u^{rs} and σ_u^{rs} respectively. Let $\Phi_{T_u^{rs}}(\cdot)$ be the cumulative distribution function (CDF) of path travel time and $\Phi_{T_u^{rs}}^{-1}(\cdot)$ be the inverse CDF of path travel time.

The most reliable path defined by Frank (1969) can be expressed with respect to the pre-specified travel time budget b as below:

Definition 1 (Most reliable path). Given a travel time budget b, a path $p_*^{rs} \in P^{rs}$ is the most reliable path if $\Phi_{T_*^{rs}}(b) > \Phi_{T_u^{rs}}(b)$ for any other path $p_u^{rs} \in P^{rs}$.

As an alternative, the α-reliable path can be defined with respect to a pre-specified on-time arrival probability α (Chen and Ji 2005) as follows:

Definition 2 (α-reliable path). Given an on-time arrival probability α, a path $p_*^{rs} \in P^{rs}$ is the α-reliable path if $\Phi_{T_*^{rs}}^{-1}(\alpha) < \Phi_{T_u^{rs}}^{-1}(\alpha)$ for any other path $p_u^{rs} \in P^{rs}$.

Travellers' different risk attitudes toward travel time uncertainty according to on-time arrival probability can be defined as below:

- If $\alpha > 0.5$ (or $\Phi_{T_*^{rs}}(b) > 0.5$) then the traveller is 'risk-averse for on-time arrival';
- If $\alpha = 0.5$ (or $\Phi_{T_*^{rs}}(b) = 0.5$) then the traveller is 'risk-neutral for on-time arrival';
- If $\alpha < 0.5$ (or $\Phi_{T_*^{rs}}(b) < 0.5$) then the traveller is 'risk-seeking for on-time arrival'.

Even though RSPP can be determined using either the most reliable path or the α-reliable path, we note that these two definitions are different to some extent. To find the most reliable path, an underlying assumption is that travellers can determine their travel time budgets prior to their trips. However, the travel time budget depends largely on the distance between O-D nodes and the congestion level in the network. The travellers may be thus unable to define an appropriate travel time budget prior to their trips. The α-reliable path definition, however, requires the travellers to express their desired risk attitudes toward travel time uncertainty. This paper postulates that in the context of pre-trip planning applications, travellers can identify their desired risk attitudes based on trip purposes rather than defining an appropriate travel time budget in the stochastic network. Therefore, the α-reliable path definition may better reflect travellers' route decision-making processes in pre-trip planning systems. In

addition, with the pre-specified travellers' risk attitude, the α-reliable path definition may lead to stricter dominance conditions and more efficient solution algorithms. These dominance conditions are discussed in the following section.

In this study, it is assumed that link travel times are non-negative and follow normal distributions; and all link travel times are statistically independent. These assumptions are commonly used in studies of stochastic shortest path problems (Fu 2001; Sen et al. 2001; Nikolova 2010). Recent empirical studies, based on field observations, also found that normal distributions appear to reflect the reality of most link travel times, and the normality assumption therefore could be sufficient from a practical standpoint given its computational simplicity (Rakha et al. 2006). Under the normality assumption, RSPP based on the α-reliable path definition can be formally defined as follows:

$$\underset{x_{ij}^{rs}}{Min} \quad \Phi_{T_*^{rs}}^{-1}(\alpha) = \sum_{a_{ij} \in A} t_{ij} x_{ij}^{rs} + Z_\alpha * \sqrt{\sum_{a_{ij} \in A} \sigma_{ij}^2 x_{ij}^{rs}} \tag{2}$$

Subject to

$$\sum_{j \in SCS(i)} x_{ij}^{rs} - \sum_{k \in PDS(i)} x_{ki}^{rs} = \begin{cases} 1 & \forall i = r \\ 0 & \forall i \neq r; i \neq s \\ -1 & \forall i = s \end{cases}, \tag{3}$$

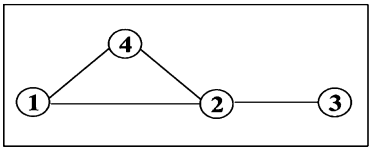
$$x_{ij}^{rs} \in \{0, 1\}, \quad \forall a_{ij} \in A \tag{4}$$

Equation (2) is the objective function that travellers want to minimize, where Z_a is the inverse CDF of the standard normal distribution at α confidence level and it can be obtained from the standard normal table or calculated by numerical approximation. Equation (3) ensures that the links on the α-reliable path are feasible. Equation (4) is concerned with the link-path incidence variables which are binary in nature.

3 Dominance conditions of RSPP

RSPP can be regarded as one of the non-additive shortest path problems (Reinhardt and Pisinger 2011), since the path cost $\left(\Phi_{T_*^{rs}}^{-1}(\alpha)\right)$ cannot be calculated as the sum of related links costs $\left(\Phi_{T_{ij}}^{-1}(\alpha)\right)$. This leads to a violation of Bellman's Principle of Optimality (Bellman 1958), which states that a sub-path between any pair of nodes on the shortest path is the shortest path itself.

Figure 1 illustrates the above non-additive property by means of a simple network. The link travel times are assumed to follow normal distribution; and the mean and variance of link travel times are given in the figure. The operation $\oplus$ is a path concatenation operator (e.g. $p_1^{13} = a_{12} \oplus a_{23}$ means that p_1^{13} goes through a_{12} and



Path	Mean	Variance	Travel time budget
a_{12}	2	2	3.81
a_{23}	3	3	4.70
$p_1^{13} = a_{12} \oplus a_{23}$	5	5	7.86
$p_2^{12} = a_{14} \oplus a_{42}$	2.5	1	3.78
$p_2^{13} = a_{14} \oplus a_{42} \oplus a_{23}$	5.5	4	8.06

Fig. 1 An illustration of non-additive property of RSPP

a_{23}). Considering a risk-averse traveller with an on-time arrival probability α=0.9 (thus Z_α=1.28), it can be seen that the cost of path p_1^{13} is not equal to the aggregate cost of a_{12} and a_{23} $\left(\Phi^{-1}_{T_1^{13}}(0.9) = 7.86 < \Phi^{-1}_{T_{12}}(0.9) + \Phi^{-1}_{T_{23}}(0.9) = 8.51\right)$. Note that the optimal path between nodes 1 and 3 uses the path p_1^{13} going through a_{12} and a_{23}. Based on Bellman's Principle of Optimality, one may expect that the optimal path from node 1 to 2 should be a_{12}, but it actually goes through $p_2^{12} = a_{14} \oplus a_{42}$.

The violation of Bellman's Principle of Optimality disallows the application of classical dynamic programming approaches (e.g. Dijkstra's algorithm) for solving RSPP. In following sections, this principle is extended and a number of dominance conditions are established for solving RSPP.

3.1 Dominance conditions between two paths

In this section, the principle of Bellman's optimality is extended and RSPP is formulated as a multi-criteria shortest path problem. The multi-criteria shortest path problem typically relies on a number of specific dominance conditions to determine dominated paths. When considering two paths $p_u^{ri} \neq p_v^{ri} \in P^{ri}$ from origin r to a node i, the following non-dominated path definition for RSPP exists:

Let $p_u^{rw} = p_u^{ri} \oplus p^{iw}$ and $p_v^{rw} = p_v^{ri} \oplus p^{iw}$ be two paths from origin r to node w going through the same sub-path p^{iw}. T_u^{rw} and T_v^{rw} denote travel time for these two paths respectively.

Definition 3 A path $p_u^{ri} \in P^{ri}$ dominates another path $p_v^{ri} \in P^{ri}$ (denoted by $p_u^{ri} \succ p_v^{ri}$), if and only if $\Phi^{-1}_{T_u^{rw}}(\alpha) < \Phi^{-1}_{T_v^{rw}}(\alpha)$ for any path $p^{iw} \in P^{iw}$ and any node $w \in N$.

Definition 4 A path $p_u^{ri} \in P^{ri}$ is a non-dominated path, if and only if p_u^{ri} is not dominated by any path $p_v^{ri} \in P^{ri}$.

Based on Definitions 3 and 4, Bellman's Principle of Optimality is extended as follows.

Theorem 1 *A sub-path of any non-dominated path must be a non-dominated path itself.*

Proof See Appendix B. □

The principle of optimality in Theorem 1 can be utilized to search for the α-reliable path using a generalized dynamic programming method. Compared with the conventional shortest path method, several non-dominated paths may have to be stored at each node in RSPP. The dominated paths at each node, however, can be discarded without further consideration since they cannot be parts of the α-reliable path between O-D nodes. To determine and discard dominated paths in the stochastic network, Miller-Hooks and Mahmassani (2003) suggested a first-order stochastic dominance (FSD) to determine the dominated paths under the most reliable path definition. As shown in Fig. 2, FSD can also be expressed under the α-reliable path definition as follows:

Proposition 1 Given two paths $p_u^{ri} \neq p_v^{ri} \in P^{ri}$, $p_u^{ri} \succ p_v^{ri}$ if p_u^{ri} and p_v^{ri} satisfy $\Phi_{T_u^{ri}}^{-1}(\lambda) < \Phi_{T_v^{ri}}^{-1}(\lambda)$ for any confidence level $0 < \lambda < 1$.

From the objective function (Eq. (2)), RSPP can be formulated as a bi-criterion shortest path problem with respect to the mean and variance of the path travel time. Sen et al. (2001) proposed a mean-variance (M-V) dominance to determine the optimal path in stochastic networks under only risk-averse scenarios. The M-V dominance can be extended for RSPP under risk-averse, risk-seeking and risk-neutral scenarios as follows:

Proposition 2 (M-V dominance). Given an on-time arrival probability *α and two paths* $p_u^{ri} \neq p_v^{ri} \in P^{ri}$, $p_u^{ri} \succ p_v^{ri}$ if p_u^{ri} and p_v^{ri} satisfy either

(i) $t_u^{ri} \leq t_v^{ri}$ and $Z_\alpha \sigma_u^{ri} < Z_\alpha \sigma_v^{ri}$ or
(ii) $t_u^{ri} < t_v^{ri}$ and $Z_\alpha \sigma_u^{ri} \leq Z_\alpha \sigma_v^{ri}$

Proof See Appendix B. □

Figure 3 graphically illustrates the M-V dominance. As shown in this figure, the M-V dominance is stronger than FSD. Using this M-V dominance, risk-averse travellers can discard dominated paths satisfying $\Phi_{T_u^{ri}}^{-1}(\lambda) < \Phi_{T_v^{ri}}^{-1}(\lambda)$ for any confidence level $0.5 \leq \lambda < 1$; and risk-seeking travellers can determine dominated paths satisfying $\Phi_{T_u^{ri}}^{-1}(\lambda) < \Phi_{T_v^{ri}}^{-1}(\lambda)$ for any confidence level $0 < \lambda \leq 0.5$.

In addition to the above two dominance conditions, Chen et al. (2008) and Hutson and Shier (2009) introduced a mean-travel time budget (M-B) dominance to

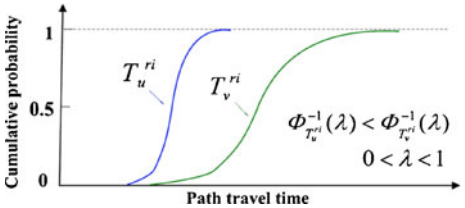


Fig. 2 First-order stochastic dominance

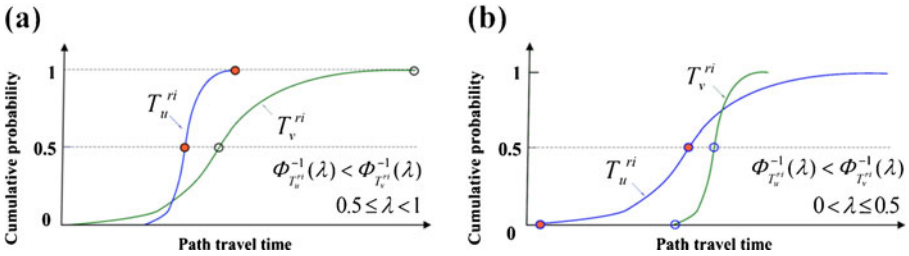


Fig. 3 Mean-variance dominance (**a**) risk-averse (**b**) risk-seeking

determine dominated paths for only risk-averse scenarios. It can be proved that the M-B dominance can also be applied to risk-seeking and risk-neutral scenarios as follows.

Proposition 3 (M-B dominance). Given an on-time arrival probability α *and two paths* $p_u^{ri} \neq p_v^{ri} \in P^{ri}$, $p_u^{ri} \succ p_v^{ri}$ if p_u^{ri} and p_v^{ri} satisfy $t_u^{ri} \le t_v^{ri}$ and $\Phi_{T_u^{ri}}^{-1}(\alpha) < \Phi_{T_v^{ri}}^{-1}(\alpha)$.

Proof See Appendix B. □

As shown in Fig. 4, the M-B dominance can help identify dominated paths which may not be identified under the M-V dominance. Using the M-B dominance, risk-averse travellers can determine and discard dominated paths that satisfy $\Phi_{T_u^{ri}}^{-1}(\lambda) < \Phi_{T_v^{ri}}^{-1}(\lambda)$ for any confidence level $0.5 \le \lambda < \alpha$. *For risk-seeking travellers, the dominated paths satisfying* $\Phi_{T_u^{ri}}^{-1}(\lambda) < \Phi_{T_v^{ri}}^{-1}(\lambda)$ for any confidence level $0.5 \le \lambda < \alpha$ *could be determined and discarded.*

Based on the above dominance conditions, it can be seen that the α-reliable path definition can lead to stricter dominance conditions than the most reliable path definition. Using the most reliable path definition, one cannot determine on-time arrival probability until the destination node has been reached. Thus, all non-dominated paths under FSD have to be kept in the path search process. The definition of the α-reliable path, however, has an explicit on-time arrival probability reflecting travellers' risk attitude toward travel time uncertainty. Such definition can lead to stricter dominance conditions (M-V dominance

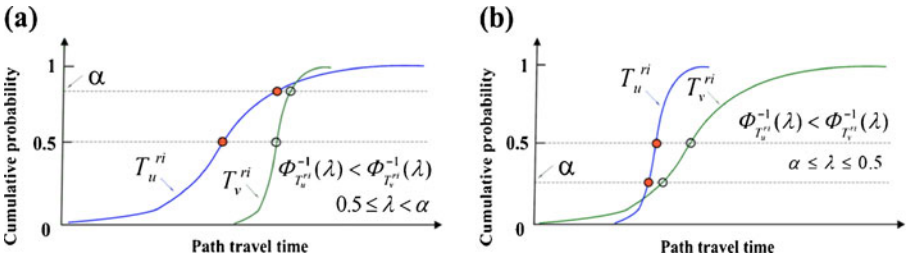


Fig. 4 Mean-travel time budget dominance (**a**) risk-averse (**b**) risk-seeking

and M-B dominance) reducing the number of generated non-dominated paths thereby improving the efficiency of the search process. In the next section, the dominance conditions, based on the α-reliable path definition, are further established for two sets of paths.

3.2 Dominance conditions between two sets of paths

Let $P^{ri} = \{p_1^{ri}, \ldots, p_n^{ri}\}$ and $\widehat{P}^{ri} = \{\widehat{p}_1^{ri}, \ldots, \widehat{p}_m^{ri}\}$ be two sets of non-dominated paths from origin r to node i. To determine their dominance relationships, a straightforward approach is to use established dominance conditions between two paths to check whether any $p_u^{ri} \in P^{ri}$ dominates any $\widehat{p}_v^{ri} \in \widehat{P}^{ri}$, and vice versa. By arranging P^{ri} and $\widehat{P}^{ri}$ as ordered sets, the following dominance conditions can be used to determine and discard a set of dominated paths in a more efficient way.

Proposition 4 Let P^{ri} be a set of non-dominated paths ordered by increasing value of t_u^{ri}. If a path $p_u^{ri} \in P^{ri}$ is ordered in front of another path $p_v^{ri} \in P^{ri} (u < v)$ then

(i) $\Phi^{-1}_{T_u^{ri}}(\lambda) \geq \Phi^{-1}_{T_v^{ri}}(\lambda)$ and
(ii) $Z_\alpha \sigma_u^{ri} \geq Z_\alpha \sigma_v^{ri}$

Proof It can be easily followed by Propositions 2 and 3. □

Given two sets of non-dominated paths P^{ri} and $\widehat{P}^{ri}$, if each path of $\widehat{P}^{ri}$ is dominated by any path of P^{ri}, $\widehat{P}^{ri}$ is then dominated by P^{ri}, denoted as $P^{ri} \succ \widehat{P}^{ri}$. In such case, all paths of $\widehat{P}^{ri}$ can be discarded without further consideration in the search process.

Proposition 5 Given two sets of non-dominated paths $P^{ri} = \{p_1^{ri}, \ldots, p_n^{ri}\}$ and $\widehat{P}^{ri} = \{\widehat{p}_1^{ri}, \ldots, \widehat{p}_m^{ri}\}$ ordered by increasing value of t_u^{ri}, then $P^{ri} \succ \widehat{P}^{ri}$ if either

(i) $t_n^{ri} \leq \widehat{t}_1^{ri}$ and $\Phi^{-1}_{T_n^{ri}}(\alpha) < \Phi^{-1}_{\hat{T}_m^{ri}}(\alpha)$ or

(ii) $t_1^{ri} \leq \widehat{t}_1^{ri}$ and $\Phi^{-1}_{T_1^{ri}}(\alpha) < \Phi^{-1}_{\hat{T}_m^{ri}}(\alpha)$

Proof (i) If $t_n^{ri} \leq \hat{t}_1^{ri}$, then $t_n^{ri} \leq \hat{t}_1^{ri} \leq \ldots \leq \hat{t}_m^{ri}$. According to Proposition 3.4(i), if $\Phi^{-1}_{T_n^{ri}}(\alpha) < \Phi^{-1}_{\hat{T}_m^{ri}}(\alpha)$, then $\Phi^{-1}_{T_n^{ri}}(\alpha) < \Phi^{-1}_{\hat{T}_m^{ri}}(\alpha) \leq \ldots \leq \Phi^{-1}_{\hat{T}_1^{ri}}(\alpha)$. Therefore, $p_n^{ri} \in P^{ri}$ dominates every path $\widehat{p}_v^{ri} \in \widehat{P}^{ri}$ according to Proposition 3.
(ii) The proof is similar to (i). □

Inspired by Skriver and Andersen (2000), the following proposition is introduced to discard all paths of $\widehat{P}^{ri}$ before the path extension operations, $\widehat{P}^{rj} = \widehat{P}^{ri} \oplus a_{ij}$, are carried out.

Proposition 6 Given two ordered sets of non-dominated paths $P^{rj} = \{p_1^{rj}, \ldots, p_n^{rj}\}$

and $\widehat{P}^{ri} = \{\widehat{p}_1^{ri}, \ldots, \widehat{p}_m^{ri}\}$, then $P^{rj} \succ \widehat{P}^{rj} = \widehat{P}^{ri} \oplus a_{ij}$ if either

(i) $t_n^{rj} \le \hat{t}_1^{ri} + t_{ij}$ and $Z_\alpha(\sigma_n^{rj})^2 < Z_\alpha(\widehat{\sigma}_m^{ri})^2 + Z_\alpha(\sigma_{ij})^2$ or

(ii) $t_1^{rj} < \hat{t}_1^{ri} + t_{ij}$ and $Z_\alpha(\sigma_1^{rj})^2 \le Z_\alpha(\widehat{\sigma}_m^{ri})^2 + Z_\alpha(\sigma_{ij})^2$

Proof (i) If $t_n^{rj} \le \hat{t}_1^{ri} + t_{ij}$, then $t_n^{rj} \le \hat{t}_1^{ri} + t_{ij} \le \ldots \le \hat{t}_m^{ri} + t_{ij}$. According to Proposition 4(ii), if $Z_\alpha(\sigma_n^{rj})^2 < Z_\alpha(\widehat{\sigma}_m^{ri})^2 + Z_\alpha(\sigma_{ij})^2$, then $Z_\alpha(\sigma_n^{rj})^2 < Z_\alpha(\widehat{\sigma}_m^{ri})^2 + Z_\alpha(\sigma_{ij})^2 \le \ldots \le Z_\alpha(\widehat{\sigma}_1^{ri})^2 + Z_\alpha(\sigma_{ij})^2$. Therefore, $p_n^{ri} \in P^{ri}$ dominates every path $\widehat{p}_i^{ri} \oplus a_{ij} \in \widehat{P}^{rj}$ according to Proposition 2.

(ii) The proof is similar to (i). □

4 Solution algorithms for solving RSPP

In this section, two efficient multi-criteria solution algorithms are presented for solving RSPP based on above established dominance conditions. Firstly, a multi-criteria label-setting algorithm, modified from a multi-criteria label-correcting shortest path algorithm (Brumbaugh-Smith and Shier 1989), is proposed in Section 4.1. Following this algorithm, a multi-criteria A* algorithm, originated from traditional A* algorithm (Hart et al. 1968), is presented in Section 4.2 to further speed up the search process.

4.1 Multi-criteria label-setting algorithm

At each node i, a set of non-dominated paths, $P^{ri} = \{p_1^{ri}, \ldots, p_n^{ri}\}$, is maintained and sorted in ascending order by mean travel time t_u^{ri}. A scan eligible set (list of nodes to be scanned), denoted by $SE = \{P^{ri}, \ldots, P^{rj}\}$, is maintained using a priority queue. All nodes in the queue are ordered based on $f(i)$, which is defined as the minimum $\Phi_{T_u^{ri}}^{-1}(\alpha)$ of all non-dominated paths in P^{ri}.

At each iteration, a node i at the top of the priority queue (with minimum $f(i)$) is selected from SE and moved to the set of selected nodes, denoted by Q. A temporary set of paths is constructed from the selected node i to each successor node j, denoted by $\widehat{P}^{rj} = P^{ri} \oplus a_{ij}$. The non-dominated paths of $\widehat{P}^{rj}$ are then merged into P^{rj} at node j, if the paths of $\widehat{P}^{rj}$ are not dominated by any path of P^{rj}. The newly generated dominated paths from $\widehat{P}^{rj}$ may also dominate existing paths in P^{rj} which should be discarded. As the algorithm progresses, each node is updated in such a way that all paths at each node are always non-dominated paths (according to the *Update* procedure). The algorithm continues the path extension process until the destination is reached or SE becomes empty. When the algorithm terminates, the α-reliable path can be determined (the last path in P^{rs} according to Proposition 4). The steps of the proposed multi-criteria label-setting algorithm, named *RSPP-LS*, are described below.

Algorithm *RSPP-LS*
Inputs: O-D pair, On-time arrival probability α
Returns: the α-reliable path
Step 1. Initialization:
Create a path p^{rr} from r to itself and set $P^{rr} := \{p^{rr}\}$.
Set $SE := \{P^{rr}\}$ and $Q := \phi$.
Step 2. Node selection:
If $SE = \phi$, stop; otherwise, continue.
Select P^{ri} at the top of SE.
Set $SE := SE \setminus \{P^{ri}\}$ and $Q := Q \cup \{P^{ri}\}$.
If $i = s$, stop; otherwise, continue.
Step 3. Path extension:
For each successor node $j \in SCS(i)$
Call procedure $P_N^{rj} := Update(P^{ri}, a_{ij}, P^{rj})$.
If $P^{rj} \notin SE$ and $P^{rj} \notin Q$, calculate $f(j)$ and set $SE := SE \cup \{P^{rj}\}$.
If $P^{rj} \in SE$, calculate $f(j)$. If $f(j)$ decreased, update the order of P^{rj} in SE.
If $P^{rj} \in Q$ and $P_N^{rj} \neq \phi$, call procedure $Extend\ (P_N^{rj})$.
End for
Go back to Step 2.

Procedure *Update*
Inputs: A set of non-dominated paths P^{ri} at selected node i, a set of existing non-dominated paths P^{rj} at successor node j, and the link a_{ij}
Returns: A set of newly generated non-dominated paths P_N^{rj} at node j
Step 1. Fast dominance check based on Proposition 6:
If $t_n^{rj} \le t_1^{ri} + t_{ij}$ and $Z_\alpha(\sigma_n^{rj})^2 < Z_\alpha(\hat{\sigma}_m^{ri})^2 + Z_\alpha(\sigma_{ij})^2$, return $P_N^{rj} := \phi$.
If $t_1^{rj} \le t_1^{ri} + t_{ij}$ and $Z_\alpha(\sigma_1^{rj})^2 < Z_\alpha(\sigma_m^{ri})^2 + Z_\alpha(\sigma_{ij})^2$, return $P_N^{rj} := \phi$.
Step 2. Path extension operation:
Set $\hat{p}_1^{rj} := p_1^{ri} \oplus a_{ij}$ and $f(j) := \Phi_{\hat{T}_1^{rj}}^{-1}(\alpha)$.
For each $p_u^{ri} \in P^{ri}$
Set $\hat{p}_u^{rj} := p_u^{ri} \oplus a_{ij}$.
If $\Phi_{\hat{T}_u^{rj}}^{-1}(\alpha) \le f(j)$, set $f(j) := \Phi_{\hat{T}_u^{rj}}^{-1}(\alpha)$ and $\hat{P}^{rj} := \hat{P}^{rj} \cup \{\hat{p}_u^{rj}\}$.
End for
Step 3. Fast dominance check based on Proposition 5:
If $t_n^{rj} \le \hat{t}_1^{rj}$ and $\Phi_{T_n^{rj}}^{-1}(\alpha) < \Phi_{\hat{T}_m^{rj}}^{-1}(\alpha)$, return $P_N^{rj} := \phi$.
If $t_1^{rj} \le \hat{t}_1^{rj}$ and $\Phi_{T_1^{rj}}^{-1}(\alpha) < \Phi_{\hat{T}_m^{rj}}^{-1}(\alpha)$, return $P_N^{rj} := \phi$.
If $\hat{t}_m^{rj} \le t_1^{rj}$ and $\Phi_{\hat{T}_m^{rj}}^{-1}(\alpha) < \Phi_{T_n^{rj}}^{-1}(\alpha)$, set $P^{rj} := \hat{P}^{rj}$ and return $P_N^{rj} := \hat{P}^{rj}$.
If $\hat{t}_1^{rj} \le t_1^{rj}$ and $\Phi_{\hat{T}_1^{rj}}^{-1}(\alpha) < \Phi_{T_n^{rj}}^{-1}(\alpha)$, set $P^{rj} := \hat{P}^{rj}$ and return $P_N^{rj} := \hat{P}^{rj}$.
Step 4. Merge operation:
Call procedure $P_N^{rj} := ModifiedMerge(P^{rj}, \hat{P}^{rj})$ and return P_N^{rj}.

Procedure *Extend*
Inputs: a set of non-dominated paths P_N^{rj} at node j
For each successor node $w \in SCS(j)$
Call procedure $P_N^{rw} := Update(P_N^{rj}, a_{jw}, P^{rw})$.
If $P^{rw} \in Q$ and $P_N^{rw} \neq \phi$, call procedure $Extend(P_N^{rw})$.
End for

The following is a description of the *Update* procedure. In Step 1, a fast dominance check is carried out using Proposition 6 before the extension and merging of operations. If Proposition 6 is satisfied, all paths in P^{ri} at selected node i are dominated and can be discarded. A temporary set of paths, $\widehat{P}^{rj} = P^{ri} \oplus a_{ij}$, will then be constructed in Step 2. Note that, in this path extension step, some paths may be dominated by the paths ahead of them, according to Proposition 4(i). In Step 3, there still exists an opportunity to discard a set of dominated paths using Proposition 5 before the merge operation is carried out. In the final step, the merge operation is carried out to combine all newly generated paths in $\widehat{P}^{rj}$ and existing paths in P^{rj}. The merge operation returns a set of newly generated non-dominated paths in $\widehat{P}^{rj}$, denoted as P_N^{rj} . The merge operation is implemented using the "modified merge algorithm" in Brumbaugh-Smith and Shier (1989).

In the conventional label-setting algorithm (Dijkstra's algorithm), paths in the selected queue Q are not updated and extended. In the proposed algorithm, newly generated non-dominated paths (P_N^{rj} return from the *Update* procedure) may need to be inserted into Q and extended to its successor nodes according to the *Extend* procedure. An example illustrating the *Extend* procedure can be found in Fig. 5. As shown in the figure, Node 1 and Node 2 (P^{11} and P^{12}) have been selected from *SE* and a non-dominated path a_{12} has been identified at Node 2. In the following iteration, Node 4 (P^{14}) is selected from *SE*, and a new non-dominated path at Node 2, $a_{14} \oplus a_{42}$, is generated. The generated non-dominated path, $a_{14} \oplus a_{42}$, should then be inserted into the set of non-dominated paths at Node 2 in Q, and further extended to Node 3, because it may result in a non-dominated path $a_{14} \oplus a_{42} \oplus a_{23}$ at Node 3.

It can be proved that the objective value $\Phi^{-1}_{T_u^{ri}}(\alpha)$ monotonically increases with path extensions as follows:

Proposition 7 Given two paths p_u^{ri} and $p_u^{rj} = p_u^{ri} \oplus a_{ij}$, the relationship $\Phi^{-1}_{T_u^{rj}}(\lambda) > \Phi^{-1}_{T_u^{ri}}(\lambda), \forall \lambda \in (0,1)$ always holds.

Proof See Appendix B. □

Based on the above monotonic property, the optimality of the *RSPP-LS* algorithm can be proved as below:

Proposition 8 The *RSPP-LS* algorithm can determine the optimal path p_*^{rs} between O-D nodes when the destination s has been selected from *SE*.

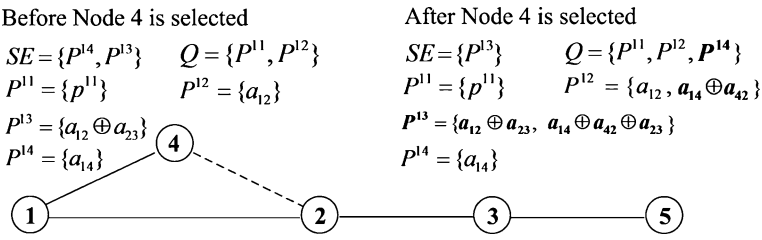


Fig. 5 An illustration of update operation

Proof Suppose that $\overline{P}^{rs}$ containing all non-dominated paths between O-D nodes, and P'^{rs} is the set of non-dominated paths when the destination s is selected from SE. As P^{rs} is selected from SE, $f(s)$ of P^{rs} is the minimum $f(i)$ of SE. Since all paths in $P'^{rs} = \overline{P}^{rs} - P^{rs}$ are extended from SE, $f(s)$ of P^{rs} is less than any $\Phi^{-1}_{T^{rs}_u}(\alpha)$ in P'^{rs}. Therefore, $f(s)$ of P^{rs} is the minimum $\Phi^{-1}_{T^{rs}_u}(\alpha)$ in $\overline{P}^{rs}$ and p^{rs}_* can be determined. □

In the worst case, the node selection step (Step 2) requires $O(|N|\log|N|)$ with the implementation of SE using an F-heap data structure (Fredman and Tarjan 1987), where $|N|$ and $|A|$ are the respective number of network nodes and links. As *Update* procedure runs in $O(|P|)$, the path extension step (Step 3) requires $O(|A||P|)$, where $|P|$ is maximum number of non-dominated paths at the nodes. Therefore, the *RSPP-LS* algorithm runs in a time $O(|A||P| + |N|\log|N|)$. Theoretically, the *RSPP-LS* algorithm has a non-polynomial complexity, because $|P|$ grows exponentially with the network size. However, a number of authors in their numerical experiments (Brumbaugh-Smith and Shier 1989; Nie and Wu 2009) found that $|P|$ is much smaller than the maximum possible size especially for sparse networks in transportation applications.

4.2 Multi-criteria A* algorithm

In this section, a multi-criteria A* algorithm, named *RSPP-A**, is proposed to further speed up the search process by assigning a high priority to those nodes closer to the destination. Denote $h(i)$ as an estimated lower bound to the actual $\Phi^{-1}_{T^{is}_u}(\alpha)$ of the optimal path p^{is}_u from node i to destination s. Note that $h(s) = 0$ at the destination. The *RSPP-A** algorithm makes use of a heuristic function $F(i) = f(i) + h(i)$ as the disutility cost of each node i instead of $f(i)$ used in the *RSPP-LS* algorithm. The value of $F(i)$ reflects the likelihood of node i being on the α-reliable path. The lower the value of $F(i)$, the more likely it is that the optimal path will go through this node. Based on this idea, the *RSPP-A** algorithm maintains an ordered SE based on the $F(i)$ value instead of $f(i)$, to determine the path extension order.

As for the conventional A* algorithm, the heuristic function in *RSPP-A** is admissible if the following inequality is satisfied

$$F\left(P^{ri} \oplus a_{ij}\right) + h(j) \geq F\left(P^{ri}\right) + h(i) \tag{5}$$

Equation (5) indicates that $F(i)$ ($F(P^{ri})$) should monotonically increase with path extensions. A common admissible $h(i)$ is the Euclidean heuristic function $d_{is}/v_{\max}$, where d_{is} is the Euclidean distance from node i to the destination s, and $v_{\max}$ is the maximum travel speed observed in the network. In practice, $v_{\max}$ can be set as the maximum speed of historical observations or simply set as the maximum design speed for the network.

Proposition 9 The *RSPP-A** algorithm can determine the optimal path p^{rs}_* between O-D nodes when the destination s is selected from SE, if the heuristic function is admissible.

Proof Suppose that $\overline{P}^{rs}$ containing all non-dominated paths between O-D nodes, and P^{rs} is the set of non-dominated paths when the destination s is selected from SE. Let $P'^{rs} = \overline{P}^{rs} - P^{rs}$. If the heuristic function is admissible, the value of selected $F(i)$ from SE monotonically increases with path extensions. As P^{rs} is selected from SE, $F(s)$ of P^{rs} is the minimum value with respect to $F(i)$ in SE. Because all paths in P'^{rs} are extended from SE, and the $F(s)$ of P^{rs} is less than the that of P'^{rs}. As $h(s) = 0$, $F(s)$ of P^{rs} is the minimum $\Phi^{-1}_{T^{rs}_u}(\alpha)$ of $\overline{P}^{rs}$, and thus p^{rs}_* can be determined. □

The actual performance of the *RSPP-A** algorithm depends on the quality of $h(i)$. Generally, the *RSPP-A** algorithm is faster than the *RSPP-LS* algorithm. In the worst case, if $h(i) = 0$, the *RSPP-A** algorithm reduces to the *RSPP-LS* algorithm.

5 Case study in Hong Kong

5.1 Data collection

A case study using real-world ATIS in Hong Kong is presented in this section to demonstrate the features of the proposed solution algorithms. In Hong Kong, a Real-time Traveller Information System (RTIS) has been developed to inform drivers estimated travel times (Tam and Lam 2008) (http://tis.td.gov.hk/rtis/ttis/index/main_partial.jsp). This system is based on real-time traffic information and offline travel data generated by a traffic flow simulator (Lam et al. 2002). The mean travel times of the links together with their variances for major urban roads are given in every 5 min. As shown in Fig. 6, the road network of the whole territory of Hong Kong has 1,367 nodes and

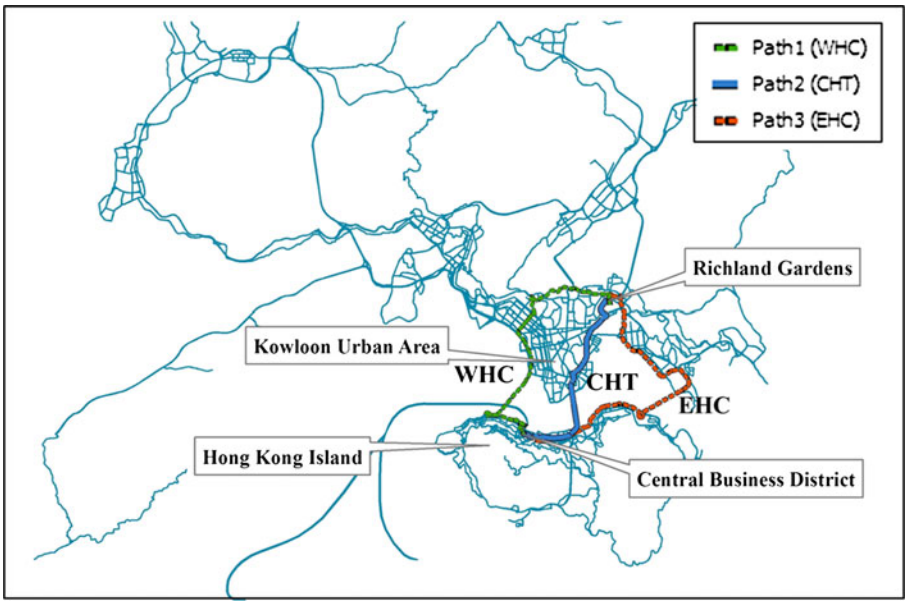


Fig. 6 RTIS network

3,655 links incorporated in RTIS. RTIS travel time data during a morning peak hour (08:00–09:00) on a typical weekday of 26 May 2006 (Friday) was collected in this case study. The toll rate data in 2006 from the Transport Department of Hong Kong Government were also collected.

To incorporate the effects of travel cost on path finding, the toll charge is incorporated into the objective function of RSPP. Suppose that travel cost associated with each link a_{ij} is a deterministic variable, denoted by τ_{ij}. The path travel cost τ_u^{rs} is the sum of link travel costs along the path p_u^{rs} as

$$\tau_u^{rs} = \sum_{a_{ij} \in A} \tau_{ij} x_{ij}^{rs} \tag{6}$$

Incorporating the travel cost into Eq. (2), the objective function becomes

$$g_u^{rs} = \tau_u^{rs} / VOT + t_u^{rs} + Z_\alpha \sigma_u^{rs} \tag{7}$$

where VOT is the value of time used for converting travel costs into a time unit equivalent. The simplest way for solving this problem is to combine travel cost and mean travel time as a new variable, denoted by $\pi_u^{rs} = \tau_u^{rs} / VOT + t_u^{rs}$, and then to formulate the objective function as

$$g_u^{rs} = \pi_u^{rs} + Z_\alpha \sigma_u^{rs} \tag{8}$$

By replacing t_u^{rs} and $\Phi_{T_u^{rs}}^{-1}(\alpha)$ by π_u^{rs} and g_u^{rs}, the *RSPP-LS* and *RSPP-A** algorithms can be directly employed to solve RSPP so as to consider travellers' various preference toward travel cost, expected travel time and travel time variation.

5.2 Numerical examples

Crossing harbour travel is one of the most important traffic patterns in Hong Kong. As shown in Fig. 6, there are three cross-harbour tunnels in Hong Kong. These three tunnels had significant differences in terms of toll rates, mean travel times and travel time variances. Cross Harbour Tunnel (CHT) was the cheapest but had the largest travel time variation. Western Harbour Crossing (WHC) was the most reliable and the expected travel time was also the shortest. The toll charge, however, was the most expensive. The expected travel time for Eastern Harbor Crossing (EHC) was the longest, but its travel time variation was similar to that of WHC.

A typical crossing harbour journey from a residential area, Richland Gardens (RG) in Kowloon, to Central Business District (CBD) in Hong Kong Island was considered in this case study. The time value for conversion of the toll rate to the time unit was set as HK$2/Minute and preferred arrival time was set as 9 AM.

Figures 6 and 7 provide travellers' departure time and route choices under various levels of on-time arrival probability (α). As shown in Fig. 6, three paths were found under various levels of on-time arrival probability (α). Path2 passing through CHT was the cheapest path (HK$ 20) but the one with the highest travel time variability ($t_2^{rs} = 36.5$ min and $\sigma_2^{rs} = 7.4$ min). In contrast, Path1 ($t_1^{rs} = 29.3$ min and $\sigma_1^{rs} = 1.2$ min) and Path3 ($t_3^{rs} = 33.1$ min and $\sigma_3^{rs} = 3.9$ min) were faster and more reliable but more expensive (HK$ 40 for Path1 and HK$ 25 for Path3).

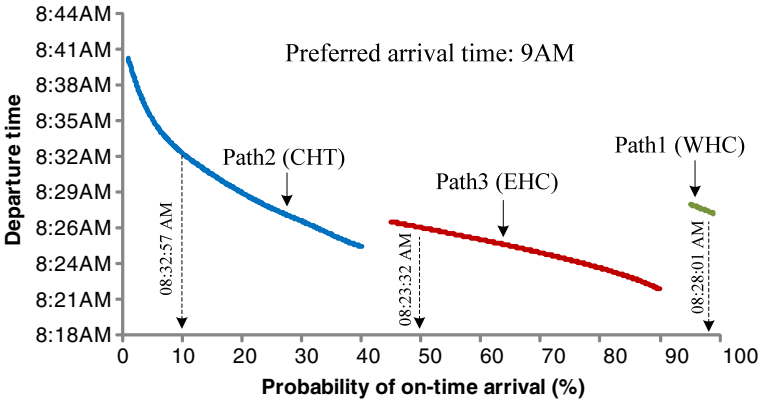


Fig. 7 Departure time and route choices under various levels of on-time arrival probability

From Fig. 7, it is seen that risk-seeking travellers (α<50 %) preferred to use Path2 passing through CHT which was the cheapest path (HK$ 20), but the one with the highest travel time variability. In this case, travellers have assigned a small budget for their travel. For example, when α=10 %, travellers can start their journeys at 8:32:57 AM with a time budget of 19.46 min. As shown in Fig. 7, risk-neutral travellers (α=50 %) prefer to choose Path3 passing EHC and use mean travel time 33.12 min as their time budget. With an increase in the on-time arrival probability, risk-averse travellers became more concerned about the travel time variability, and consequently assign a larger travel time budget by departing from their origin node earlier. For example, when α=50 %, travellers started their trips at 8:23:32 AM; when α=90 % travellers depart 5.01 min earlier. In addition to the adjustment of departure time choice, travellers may change their route choice to obtain a higher probability of on-time arrival. For example, risk-averse can switch to Path1 passing WHC which was faster and more reliable but more expensive. In this case, travellers can depart later from their origin at 8:28:01 AM, and achieve a high probability of on-time arrival α=99 %.

In this case study, the α-reliable paths were further computed to the same CBD for 333 O-D nodes from all zone centres. The tunnel choice proportions were studied to demonstrate the global features of travellers' route choices. Figure 8 illustrates the path finding results. It can be seen from the figure that travellers' route choices changed significantly for various levels of on-time arrival probability. As shown in the figure, CHT was the most frequently used, when travellers were risk-seeking. However, with an increase of on-time arrival probability, risk-averse travellers tended to switch to WHC leading to a significant decrease in the use of CHT. This demonstrated that travelers, more concerned with on-time arrival, would avoid those routes with high travel time variability.

6 Computational performance

This section examines the computational performance of *RSPP-LS* and *RSPP-A** algorithms on several large-scale networks. The *RSPP-LS* and *RSPP-A** algorithms

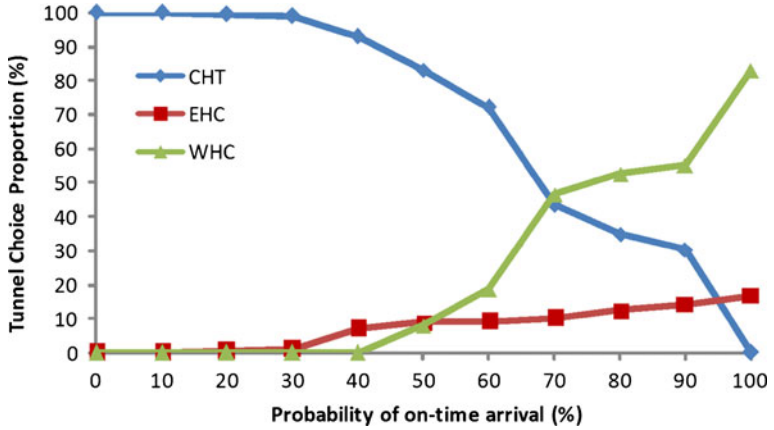


Fig. 8 Travellers' route choices from 333 zone centres to CBD

were coded in the Visual C# programming language. The priority queue was implemented using the F-heap data structure (Fredman and Tarjan 1987). To comparatively evaluate and benchmark the proposed algorithms, an exact solution algorithm for solving RSPP under risk-averse scenarios (*RSPP-N*) (Nikolova 2010) was implemented. Following the recommendations of the author, Dijkstra's algorithm (using the same F-heap data structure) was employed as the deterministic shortest path algorithm for the implementation of *RSPP-N*. As RSPP can be considered as a bi-criterion shortest path problem (BSPP) with respect to the mean travel time and its variance, the label-correcting algorithm (*BSPP-LCA*) (Skriver and Andersen 2000), which is known to be the most efficient for solving the BSPP, was also implemented for comparison purpose. These two additional solution algorithms were coded in the same environment and tested on the same networks. All experiments were conducted on ThinkPad X61 laptop with an Intel dual-core 1.6 GHz CPU (only a single processor was used) and 2 G RAM.

The computational tests were conducted on three different types of network: road networks, grid networks and random networks. NetMaker (Skriver and Andersen 2000) was used to generate random networks without the mean and variance of link travel times. To test the *RSPP-A** algorithm, the x and y coordinates of nodes were uniform distributed from [0 m, 10,000 m], respectively. Based on the node coordinates, the link lengths was calculated as the Euclidean distance between the two nodes of a link. Then, link speeds were uniformly distributed from [10 km/h, 100 km/h] to generate the mean travel time by calculating the ratio of link length to link speed. SD of link travel times was generated by randomly selecting the coefficient of variation (the ratio of SD to mean) from a uniform distribution in the range of [0.1, 1]. This method to randomly generate the mean and variance of link travel times was also adopted for the grid networks and the Chicago regional network. For the Hong Kong RTIS network, the mean and variance of link travel times were collected from RTIS as described in Section 5. The characteristics of these eight networks are given in Table 1.

Tables 2 and 3 report the computational performance of the algorithms when solving RSPP. The computational performance was evaluated in terms of

Table 1 Basic characteristics of testing networks

Road networks			Grid networks			Random networks		
Network	$\|N\|$	$\|A\|$	Network	$\|N\|$	$\|A\|$	Network	$\|N\|$	$\|A\|$
Hong Kong RTIS	1,367	3,655	G1 (40*50)	2,000	7,820	R1	3,000	24,541
			G2 (50*100)	5,000	19,700	R2	7,000	57,369
Chicago regional	12,982	39,018	G3(100*100)	10,000	39,600	R3	14,000	114,532

$|N|$: Number of nodes in the network

$|A|$: Number of links in the network

computational time (denoted by $\tilde{t}$ in milliseconds) and the number of generated non-dominated paths between O-D nodes (denoted by $\tilde{n}$). Note that $\tilde{n}$ can be interpreted as the number of extreme points in Nikolova (2010). All the reported $\tilde{t}$ and $\tilde{n}$ values were average of 100 runs, using different O-D nodes for each run. The 100 O-D nodes were randomly selected for each network and the same set of O-D nodes was used for every test performed on a given network.

Table 2 shows the results for risk-averse scenarios. The on-time arrival probability was set as α=90 %. As shown in the table, the proposed *RSPP-LS* algorithm performed substantially better than the *RSPP-N* and *BSPP-LCA* algorithms for all types of networks. For instance, *RSPP-LS* required 94.96 milliseconds to determine the α-reliable path in the Chicago regional network. This was 24.8 (2,353/94.96) times faster than *RSPP-N* and 14,387 (1,365,337/94.96) times faster than *BSPP-LCA*. These results were expected since *RSPP-LS* generated fewer non-dominated paths than the *RSPP-N* and *BSPP-LCA* algorithms.

Figure 9 illustrates the number of non-dominated paths ($\tilde{n}$) generated by different algorithms to search for the α-reliable path under risk-averse scenarios. The *BSPP-LCA* algorithm first determines all of the non-dominated paths between the O-D

Table 2 Computational performance of algorithms for risk-averse scenarios

Network	RSPP-A*		RSPP-LS		RSPP-N		BSPP-LCA	
	$\tilde{t}$	$\tilde{n}$	$\tilde{t}$	$\tilde{n}$	$\tilde{t}$	$\tilde{n}$	$\tilde{t}$	$\tilde{n}$
RTIS	3.37	1.22	4.30	1.22	67.21	10.82	2,382	34.31
Chicago	52.36	2.46	94.96	2.50	2,353	19.51	1,365,337	195.2
G1	4.84	1.72	7.92	1.77	79.63	7.48	773.33	23.3
G2	18.88	2.15	28.89	2.18	470.20	12.05	18,533	51.81
G3	40.86	2.34	81.43	2.35	1,443	15.04	78,481	76.91
R1	63.02	1.44	96.91	1.50	620.61	4.96	382.22	9.58
R2	278.32	1.66	387.06	1.75	3,792	5.09	1,488	10.06
R3	874.03	1.43	1,184	1.46	13,452	4.84	3,166	9.79

$\tilde{t}$: Average computational times in milliseconds (100 runs)

$\tilde{n}$: Average numbers of generated non-dominated paths between O-D nodes (100 runs)

Table 3 Computational performance of algorithms for risk-seeking scenarios

Network	*RSPP-A**		*RSPP-LS*	
	$\widetilde{t}$	$\widetilde{n}$	$\widetilde{t}$	$\widetilde{n}$
Hong Kong RTIS	3.23	1.17	4.19	1.18
Chicago region	47.24	2.36	93.83	2.52
G1	4.18	1.74	7.88	1.97
G2	16.77	1.92	26.86	1.96
G3	36.44	1.95	76.10	2.05
R1	50.79	1.54	95.27	1.81
R2	216.24	1.58	362.12	1.77
R3	655.88	1.19	1,142.20	1.38

$\widetilde{t}$: Average computational times in milliseconds (100 runs)

$\widetilde{n}$: Average numbers of generated non-dominated paths between O-D nodes (100 runs)

nodes (all of the points in the Fig. 9) ($\widetilde{n} = 195.2$ for the Chicago regional network), and then chose the α-reliable path with minimum travel time budget. As RSPP in risk-averse scenarios can be formulated as a convex combination of mean travel time and its variance, the *RSPP-N* algorithm only determined the non-dominated paths on the convex hull ($\widetilde{n} = 19.51$ for Chicago regional network) without consideration of non-dominated paths in the shaded areas shown in Fig. 9. Each non-dominated path on the convex hull was obtained by a search of the shortest path in a deterministic network using Dijkstras' algorithm. The proposed *RSPP-LS* algorithm, however, can determine the α-reliable path as soon as the destination node was reached. Thus, only a few non-dominated paths ($\widetilde{n} = 2.5$ for Chicago regional network) were generated. Therefore, the proposed *RSPP-LS* algorithm performed better than the other two algorithms, especially for networks with a large number of non-dominated paths.

It can be observed from Table 2 that the *RSPP-A** algorithm significantly reduces the time searching for the α-reliable path when compared to the performance of the *RSPP-LS* algorithm. For example, *RSPP-A** can reduce computational time by 21.7 % for the RTIS network of Hong Kong; and 49.8 % for the grid network of G3. These results were mainly due to the fact that the *RSPP-A** algorithm tended to search among a considerably smaller set of nodes than that of the *RSPP-LS* algorithm.

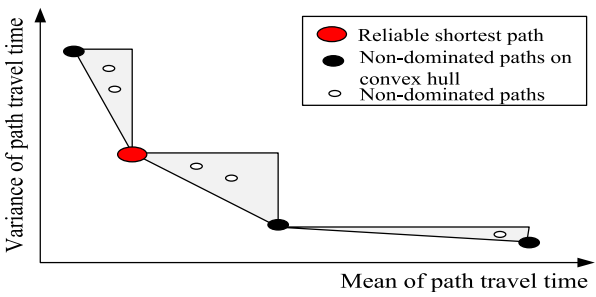


Fig. 9 Number of non-dominated paths generated by three different algorithms

The fact of exploring smaller sets of nodes may also contribute to the generation of fewer non-dominated paths ($\widetilde{n}$) during the path searching process (see Table 2).

It can also be seen from Table 2 that the *RSPP-N* algorithm result was superior to that of the *BSPP-LCA* algorithm for road networks and grid networks, where the number of non-dominated paths is large and link density is small ($|A|/|N|$ was about 3 to 4). However, the performance of *RSPP-N* may even become inferior to *BSPP-LCA* for random networks with a few non-dominated paths and a large density ($|A|/|N|$ was about 8). It should be noted that *RSPP-A** was still the best performer even for these random networks.

Table 3 reports the results for solving RSPP under risk-seeking scenarios (α= 10 %). Under risk-seeking scenarios, the *RSPP-N* and *BSPP-LCA* algorithms cannot be used to solve RSPP. This is because risk-seeking travellers tended to choose optimal paths with smaller means but larger variances. In this case, the α-reliable path is not guaranteed on the convex hull and thus the *RSPP-N* algorithm cannot be used for solving the problem. In addition, when travelers are risk-seeking, the non-dominated paths may contain an infinite number of cycles under the M-V dominance. As a result, the *BSPP-LCA* algorithm is computationally intractable for solving RSPP under risk-seeking scenarios. In contrast to these two algorithms, the *RSPP-LS* and *RSPP-A** algorithm based on M-B dominance can solve RSPP in the same way for both risk-averse and risk-seeking scenarios. Therefore, the proposed *RSPP-LS* and *RSPP-A** can determine the α-reliable path for risk-seeking travellers as efficiently as for the risk-averse travellers (see Table 3).

7 Conclusions

The reliable shortest path problem (RSPP) has been investigated using the definition of α-reliable path, which minimizes the travel time budget required to ensure a certain level of on-time arrival probability. Several dominance conditions were established based on the α-reliable path definition. These established dominance conditions can reduce the number of non-dominated paths generated, evaluated, and stored when searching for the α-reliable path. Built on the established dominance conditions, two multi-criteria solution algorithms were proposed to solve RSPP. The proposed multi-criteria label-setting algorithm, named *RSPP-LS*, can determine the optimal path without requirement of generating all non-dominated paths in the network. The multi-criteria A* algorithm, named *RSPP-A**, can speed up the search process by assigning priority to the nodes closer to the destination.

The performance of *RSPP-LS* and *RSPP-A** algorithms was examined on six large-scale networks. The computational results showed that the proposed *RSPP-LS* and *RSPP-A** algorithms performed substantially better than previous algorithms (Nikolova 2010; Skriver and Andersen 2000) under risk-averse scenarios. For the risk-seeking scenarios (which cannot be solved by previous algorithms), the proposed *RSPP-LS* and *RSPP-A** algorithms were able to solve the problem as efficiently as for the risk-averse scenarios. The computational results also indicate that the *RSPP-A** algorithm can significantly reduce computational time by about 20–50 % when compared to the *RSPP-LS* algorithm. It is evident that the proposed algorithms can determine the α-reliable path in large-scale networks in a satisfactory computation

time. This confirms that the proposed solution algorithms are applicable to online pre-trip planning system in large-scale networks. A comprehensive numerical example was presented using the Real-time Traveller Information System (RTIS) in Hong Kong. Experimental results indicated that the proposed solution algorithms are sufficiently robust to take account of the various examples of travellers' risk-taking behaviour, in terms of departure time choices and route choices.

On the basis of the algorithm proposed in this paper, several further extensions can be envisaged. (1) In this study, link travel times are assumed to follow a normal distribution. However, the normal distribution of link travel time may not always hold in reality. How to incorporate other travel time distributions, such as the log-normal distribution, in the proposed reliable shortest algorithms needs further investigations. (2) It should be pointed out that the correlations of link travel times are omitted in this study for simplicity. Recent empirical studies have found that travel times are strongly correlated among neighboring links (Chan et al. 2009). Chen et al. (2012b) extends the proposed solution algorithms to integrate such travel time correlations. (3) It is assumed in this study that link travel times are stable for a relatively long time period (e.g. morning peak hours). Further investigation is warranted for extending the proposed algorithms to stochastic time-dependent network where link travel times vary with the time-of-day (Chen et al. 2012c). Another research direction is to extend the proposed algorithms for in-vehicle navigation applications, in which the en-route traffic information is updated continuously during journeys (Fu 2001; Waller and Ziliaskopoulos 2002; Fan and Nie 2006).

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Appendix A. Notations

A	A set of links
$\|A\|$	Number of links in the network
a_{ij}	A link connecting node i and node j
b	Travel time budget
d_{is}	Euclidean distance from node i to destination
$F(i)$	Heuristic value function for the path set P^{rs}
$f(i)$	Minimum travel time budget of the path set P^{rs}
G	Directed network
$h(i)$	Estimation of travel time budget from node i to destination
N	A set of nodes
$\|N\|$	Number of nodes in the network
P^{rs}	A set of paths from origin to destination
$\|P\|$	Number of non-dominated paths between origin and destination
p_u^{rs}	A path from origin to destination

p_*^{rs}	Optimal path from origin to destination
$PDS(i)$	A set of predecessor nodes of node i
r	Origin node
s	Destination node
$SCS(i)$	A set of successor nodes of node i
T_{ij}	Travel time distribution of link a_{ij}
t_{ij}	Mean travel time of link a_{ij}
T_u^{rs}	Travel time distribution of path p_u^{rs}
t_u^{rs}	Mean travel time of path p_u^{rs}
$\nu_{\max}$	Maximum travel speed in the network
VOT	Value of time
Z_α	Inverse cumulative distribution function for standard normal distribution at α confidence level
α	Probability of on-time arrival
σ_{ij}	Travel time standard deviation of link a_{ij}
σ_u^{rs}	Travel time standard deviation of path p_u^{rs}
x_{ij}^{rs}	Link-path incidence relationship
τ_u^{rs}	Toll charge of path p_u^{rs}
τ_{ij}	Toll charge of link a_{ij}
$f_{T_u^{rs}}(\cdot)$	Probability density function of path travel time distribution
$\Phi_{T_u^{rs}}(\cdot)$	Cumulative distribution function of path travel time distribution
$\Phi_{T_u^{rs}}^{-1}(\cdot)$	Inverse cumulative distribution function of path travel Time distribution

Appendix B. Proofs

Theorem 1 A sub-path of any non-dominated path must be a non-dominated path itself.

Proof Suppose $p_v^{ri} \in P^{ri}$ is a sub-path of a non-dominated path $p_v^{rw} = p_v^{ri} \oplus p_v^{iw} \in P^{rw}$, and it is dominated by another path $p_u^{ri} \in P^{ri}$. According to Definition 3.3, we have $\Phi_{T_u^{rs}}^{-1}(\alpha) < \Phi_{T_v^{rs}}^{-1}(\alpha)$ for any path $p^{is} \in P^{is}$. Therefore, there exists at least one path $p_u^{rw} = p_u^{ri} \oplus p_v^{iw} \in P^{rw}$ satisfying $\Phi_{T_u^{rs}}^{-1}(\alpha) < \Phi_{T_v^{rs}}^{-1}(\alpha)$ for any $p^{ws} \in P^{ws}$, since any $p_v^{iw} \oplus p^{ws} \in P^{is}$. Thus, $p_u^{ri} \succ p_v^{ri}$. This contradicts the fact that p_v^{ri} is a non-dominated path. □

Let $p_u^{rw} = p_u^{ri} \oplus p^{iw}$ and $p_v^{rw} = p_v^{ri} \oplus p^{iw}$ be two paths from origin r to node w going through the same sub-path p^{iw}. T_u^{rw} and T_v^{rw} denote travel time for these two paths respectively. We introduce following function to facilitate the proofs of Propositions 2 and 3.

$$g_{uv}^{ri}(p^{iw}) = \Phi_{T_u^{rw}}^{-1}(\alpha) - \Phi_{T_v^{rw}}^{-1}(\alpha) = \left(t_u^{ri} - t_v^{ri}\right) + Z_\alpha\left(\sqrt{\left(\sigma_u^{ri}\right)^2 + \left(\sigma^{iw}\right)^2} - \sqrt{\left(\sigma_v^{ri}\right)^2 + \left(\sigma^{iw}\right)^2}\right) \tag{9}$$

The $g_{uv}^{ri}(p^{iw})$ has following properties:

Lemma 1 (i) If $Z_\alpha \sigma_u^{ri} < Z_\alpha \sigma_v^{ri}$, and thus $g_{uv}^{ri}(p^{iw})$ is a monotonic increasing function, $g_{uv}^{ri}(p^{iw}) \in \left[\Phi_{T_u^{ri}}^{-1}(\alpha) - \Phi_{T_v^{ri}}^{-1}(\alpha), t_u^{ri} - t_v^{ri}\right)$;

(ii) If $Z_\alpha\sigma_u^{ri} > Z_\alpha\sigma_v^{ri}$, and thus $g_{uv}^{ri}(p^{iw})$ is a monotonic decreasing function, $g_{uv}^{ri} \times (p^{iw}) \in \left(t_u^{ri} - t_u^{ri}, \Phi_{T_u^{ri}}^{-1}(\alpha) - \Phi_{T_v^{ri}}^{-1}(\alpha)\right]$;

(iii) If $Z_\alpha\sigma_u^{ri} = Z_\alpha\sigma_v^{ri}$, and $g_{uv}^{ri}(p^{iw}) = t_u^{ri} - t_v^{ri}$.

Proof The gradient of $g_{uv}^{ri}(p^{iw})$ can be formulated as $g_{uv}^{ri\,\prime}(p^{iw}) = \frac{\partial g_{uv}^{ri}(p^{iw})}{\partial(\sigma^{iw})^2} = \frac{Z_\alpha}{2} \times \frac{\sqrt{\left(\sigma_v^{ri}\right)^2+(\sigma^{iw})^2}-\sqrt{\left(\sigma_u^{ri}\right)^2+(\sigma^{iw})^2}}{\sqrt{\left(\sigma_u^{ri}\right)^2+(\sigma^{iw})^2}*\sqrt{\left(\sigma_v^{ri}\right)^2+(\sigma^{iw})^2}}$. Therefore, when $Z_\alpha\sigma_u^{ri} < Z_\alpha\sigma_v^{ri}$, we have $g_{uv}^{ri\,\prime}(p^{iw}) > 0$ and $\Phi_{T_u^{ri}}^{-1}(\alpha) - \Phi_{T_v^{ri}}^{-1}(\alpha) \le g_{uv}^{ri}(p^{iw}) < t_u^{ri} - t_v^{ri}$; when $Z_\alpha\sigma_u^{ri} > Z_\alpha\sigma_v^{ri}$, we have $g_{uv}^{ri\,\prime}(p^{iw}) < 0$ and $t_u^{ri} - t_v^{ri} < g_{uv}^{ri}(p^{iw}) \le \Phi_{T_u^{ri}}^{-1}(\alpha) - \Phi_{T_v^{ri}}^{-1}(\alpha)$; when $Z_\alpha\sigma_u^{ri} = Z_\alpha\sigma_v^{ri}$, we have $g_{uv}^{ri\,\prime}(p^{iw}) = 0$ and $g_{uv}^{ri}(p^{iw}) = t_u^{ri} - t_v^{ri}$.□

According to Definition 3, $p_u^{ri} \succ p_v^{ri}$ is equivalent to $g_{uv}^{ri}(p^{iw}) < 0$ for any path $p^{iw} \in P^{iw}$. We will prove $g_{uv}^{ri}(p^{iw}) < 0$ for following dominance conditions.

Proposition 2 (M-V dominance). Given an on-time arrival probability α and two paths $p_u^{ri} \ne p_v^{ri} \in P^{ri}$, $p_u^{ri} \succ p_v^{ri}$ if p_u^{ri} and p_v^{ri} satisfy either

(i) $t_u^{ri} \le t_v^{ri}$ and $Z_\alpha\sigma_u^{ri} < Z_\alpha\sigma_v^{ri}$ or
(ii) $t_u^{ri} < t_v^{ri}$ and $Z_\alpha\sigma_u^{ri} \le Z_\alpha\sigma_v^{ri}$

Proof It can be easily followed by Lemma 1.□

Proposition 3 (M-B dominance). Given an on-time arrival probability α and two paths $p_u^{ri} \ne p_v^{ri} \in P^{ri}$, $p_u^{ri} \succ p_v^{ri}$ if p_u^{ri} and p_v^{ri} satisfy $t_u^{ri} \le t_v^{ri}$ and $\Phi_{T_u^{ri}}^{-1}(\alpha) < \Phi_{T_v^{ri}}^{-1}(\alpha)$.

Proof When $Z_\alpha\sigma_u^{ri} < Z_\alpha\sigma_v^{ri}$ and $t_u^{ri} \le t_v^{ri}$ according to Lemma 1(i), we have $g_{uv}^{ri} \times (p^{iw}) < \left(t_u^{ri} - t_v^{ri}\right) \le 0$. When $Z_\alpha\sigma_u^{ri} > Z_\alpha\sigma_v^{ri}$ and $t_u^{ri} \le t_v^{ri}$ according to Lemma 1(ii), we have $g_{uv}^{ri}(p^{iw}) < \left(\Phi_{T_u^{ri}}^{-1}(\alpha) - \Phi_{T_v^{ri}}^{-1}(\alpha)\right) \le 0$. When $Z_\alpha\sigma_u^{ri} = Z_\alpha\sigma_v^{ri}$ and $\Phi_{T_u^{ri}}^{-1} \times (\alpha) < \Phi_{T_v^{ri}}^{-1}(\alpha)$, we have $t_u^{ri} < t_v^{ri}$. According to Lemma 1(iii), we have $g_{uv}^{ri} \times (p^{iw}) = \left(t_u^{ri} - t_v^{ri}\right) < 0$. Therefore, when $t_u^{ri} \le t_v^{ri}$ and $\Phi_{T_u^{ri}}^{-1}(\alpha) < \Phi_{T_v^{ri}}^{-1}(\alpha)$, we have $g_{uv}^{ri}(p^{iw}) < 0$, $\forall p^{iw} \in P^{iw}$, $\forall w \in N$. □

Let $f_{T_u^{ri}}(t)$ be the PDF of path travel time distribution T_u^{ri}. It can be proved that the objective value $\Phi_{T_u^{ri}}^{-1}(\alpha)$ monotonically increases with path extensions as follows:

Proposition 7 Given two paths p_u^{ri} and $p_u^{rj} = p_u^{ri} \oplus a_{ij}$, the relationship $\Phi_{T_u^{rj}}^{-1}(\lambda) > \Phi_{T_u^{ri}}^{-1}(\lambda), \forall\lambda \in (0,1)$ always holds.

Proof $\Phi_{T_u^{rj}}(b) - \Phi_{T_u^{ri}}(b) = \int_0^b \Phi_{T_{ij}}(b-t) f_{T_u^{ri}}(t)dt - \int_0^b f_{T_u^{ri}}(t)dt$
$= \int_0^b \left(\Phi_{T_{ij}}(b-t) - 1\right) f_{T_u^{ri}}(t)dt < 0, \quad \forall b \in R^+$
$\Rightarrow \Phi_{T_u^{rj}}^{-1}(\lambda) > \Phi_{T_u^{ri}}^{-1}(\lambda), \quad \forall\lambda \in (0,1)$.□

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